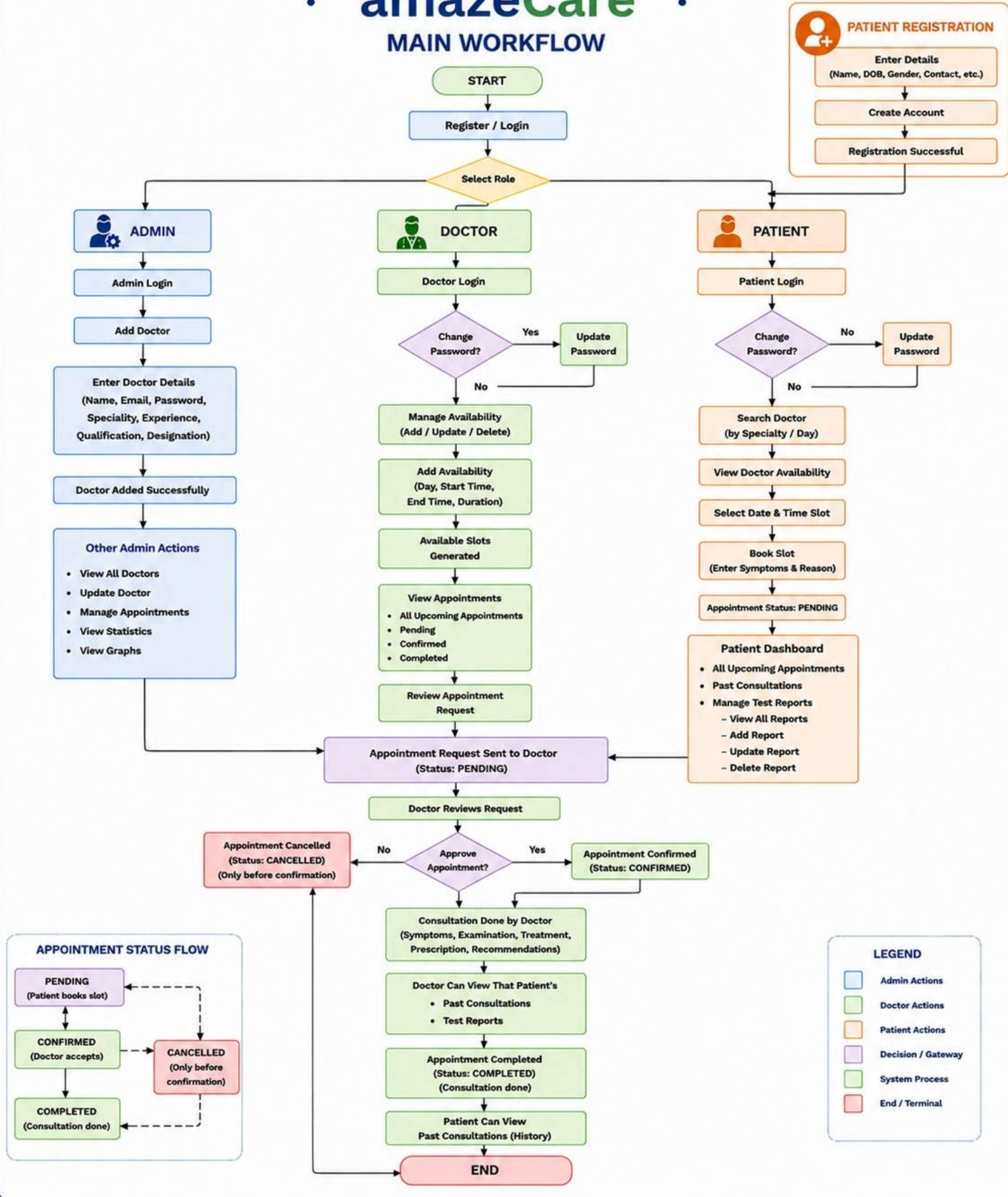
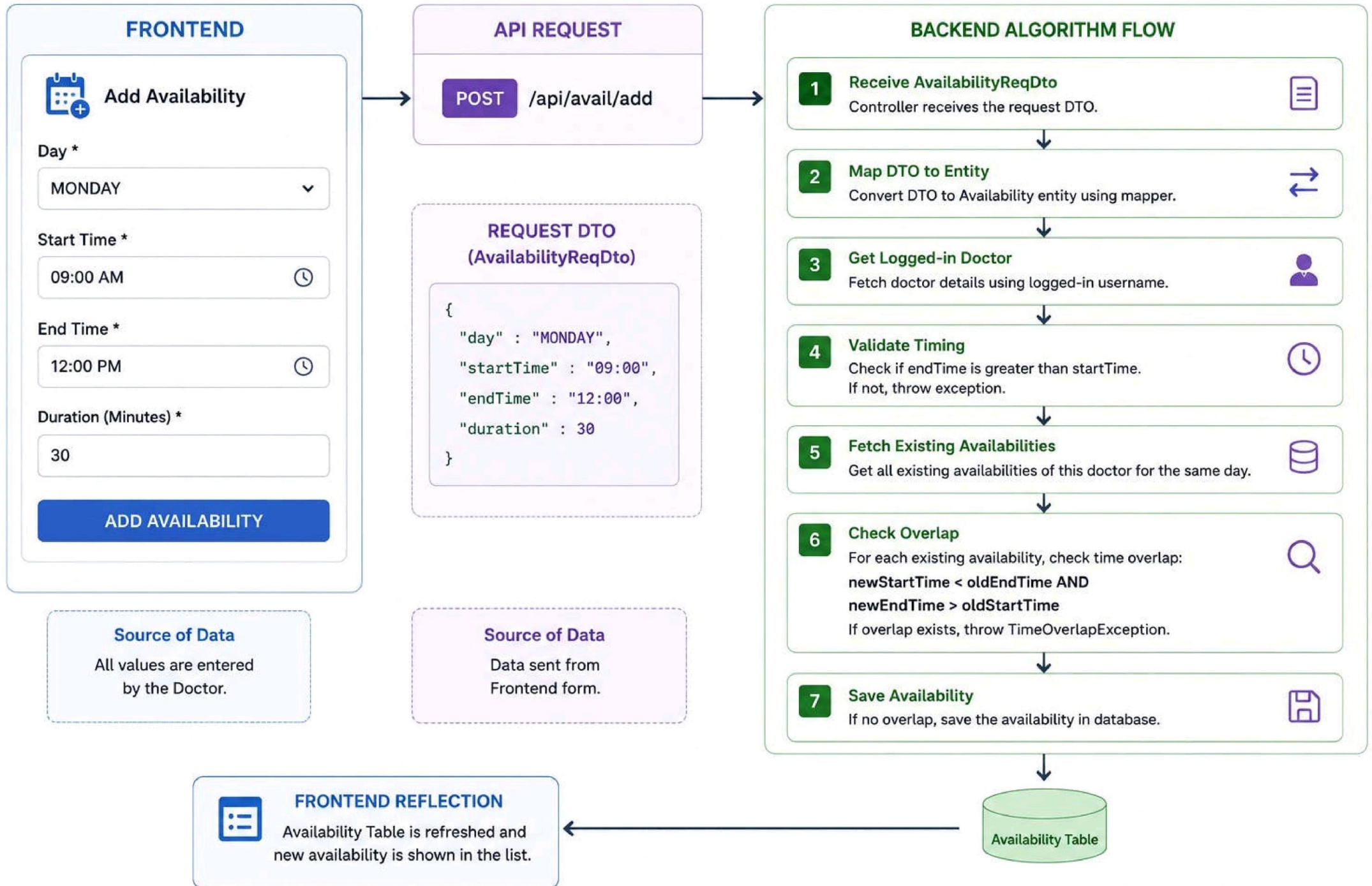


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MAIN WORKFLOW



1. ADD AVAILABILITY – ALGORITHM FLOW



2. UPDATE AVAILABILITY – ALGORITHM FLOW

FRONTEND



Availability Table

Day	Time	Duration	Actions
MONDAY	09:00 AM – 11:00 AM	30 min	
MONDAY	11:30 AM – 01:00 PM	30 min	
TUESDAY	04:00 PM – 06:00 PM	30 min	
WEDNESDAY	10:00 AM – 12:00 PM	30 min	

Doctor clicks Update



Update Availability

Day *

MONDAY

Start Time *

09:00 AM

End Time *

11:30 AM

Duration (Minutes) *

30

UPDATE AVAILABILITY

Source of Data

Existing availability data is loaded from database and shown in the table.

API REQUEST

PUT /api/avail/update/{id}

REQUEST DTO (AvailabilityReqDto)

```
{
  "day" : "MONDAY",
  "startTime" : "09:00",
  "endTime" : "11:30",
  "duration" : 30
}
```

Source of Data

Updated values are entered by the Doctor in the form.

BACKEND ALGORITHM FLOW

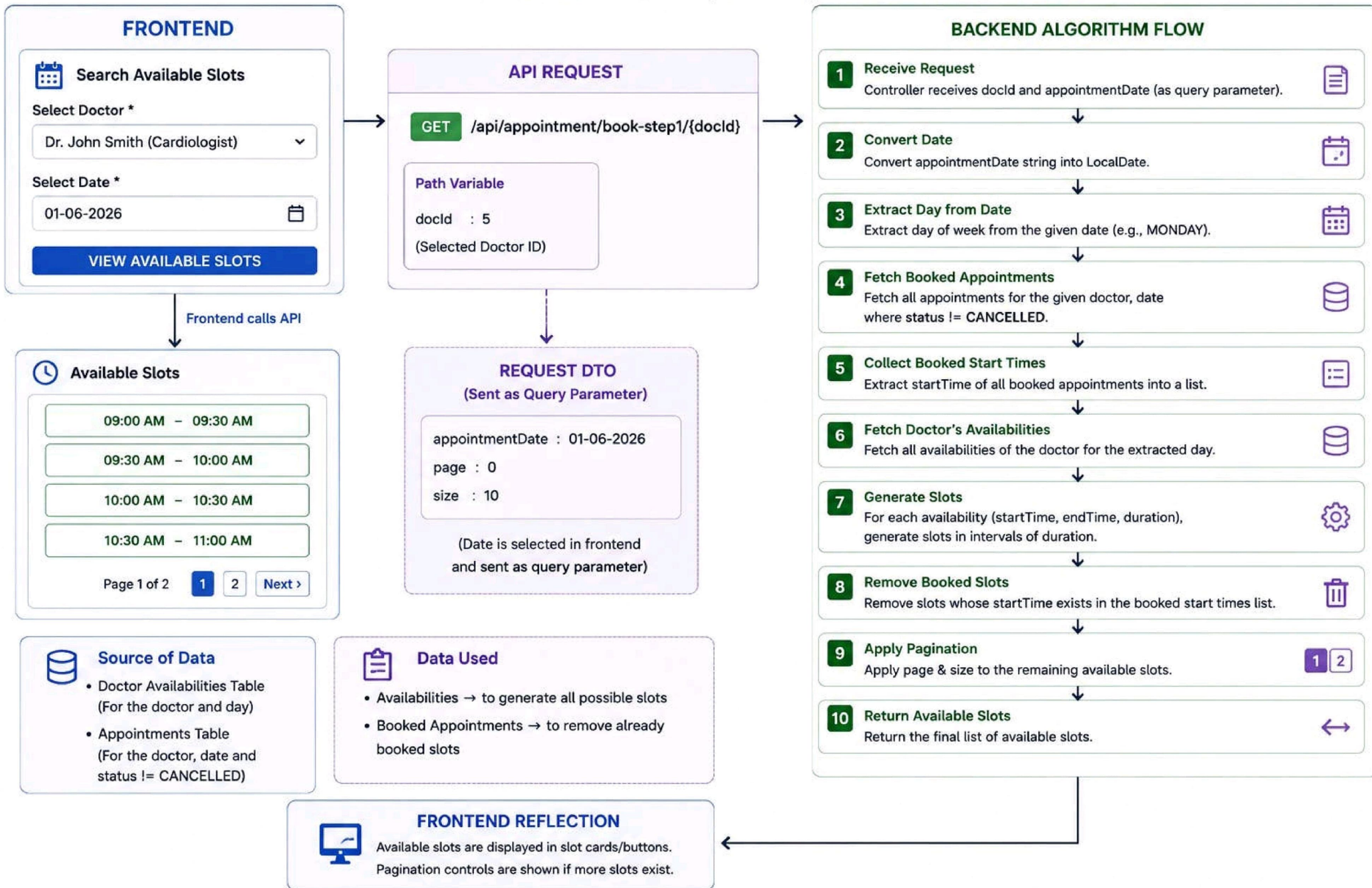
- 1 Fetch Existing Availability**
Fetch availability from database using the given ID.
- 2 Update Fields**
Update the entity fields with new values from DTO.
- 3 Validate Timing**
Check if endTime is greater than startTime.
If not, throw exception.
- 4 Fetch Doctor's Availabilities for Same Day**
Get all other availabilities of this doctor for the same day.
- 5 Ignore Current Record**
Remove the current availability (same ID) from the list before checking overlap.
- 6 Check Overlap**
For each remaining availability, check time overlap:
newStartTime < oldEndTime AND newEndTime > oldStartTime
If overlap exists, throw TimeOverlapException.
- 7 Save Updated Availability**
If no overlap, save the updated availability in database.



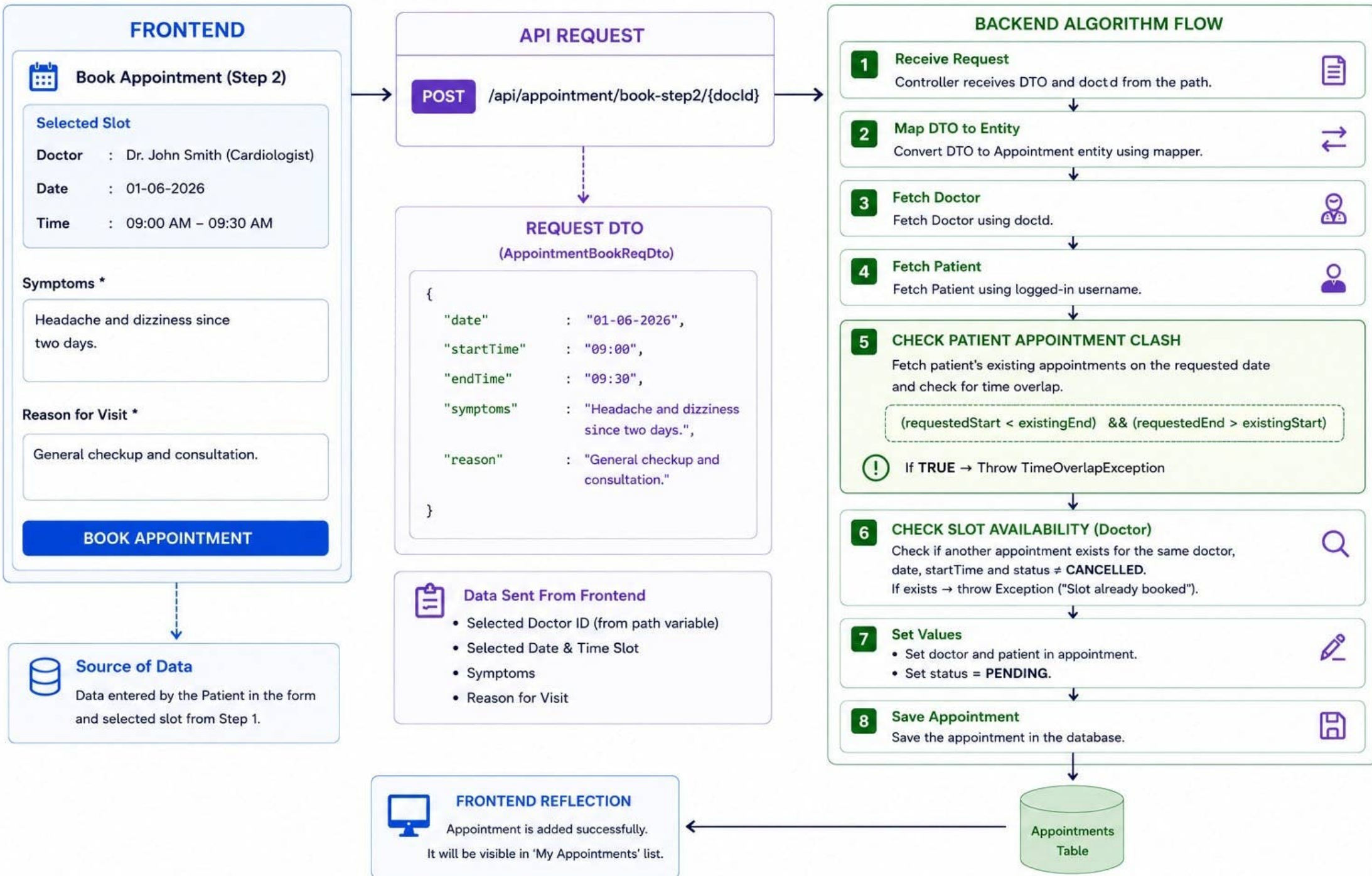
FRONTEND REFLECTION

Availability table is refreshed and updated availability is shown in the list.

3. VIEW AVAILABLE SLOTS (STEP 1) – ALGORITHM FLOW



4. BOOK APPOINTMENT (STEP 2) – ALGORITHM FLOW



5. SLOT GENERATION LOGIC – ALGORITHM FLOW

INPUT (One Availability)

 **Start Time** : 09:00 AM
 **End Time** : 11:00 AM
 **Duration** : 30 Minutes

STEP 1: Generate All Possible Slots

Generate slots in intervals of 'duration' from startTime to endTime.

1 09:00 AM - 09:30 AM
2 09:30 AM - 10:00 AM
3 10:00 AM - 10:30 AM
4 10:30 AM - 11:00 AM

(Loop Condition: $\text{currentTime} \leq \text{endTime} - \text{duration}$)

ALGORITHM STEPS

1 **Take Availability**
Read startTime, endTime, duration.

2 **Initialize Current Time**
 $\text{currentTime} = \text{startTime}$

3 **Generate Slots**
While $\text{currentTime} \leq \text{endTime} - \text{duration}$
• $\text{slotStart} = \text{currentTime}$
• $\text{slotEnd} = \text{currentTime} + \text{duration}$
• Add ($\text{slotStart} - \text{slotEnd}$) to slot list
• $\text{currentTime} = \text{currentTime} + \text{duration}$

4 **Get Booked Start Times**
Fetch all booked appointments for the doctor and date (**status** != CANCELLED).

5 **Remove Booked Slots**
Remove slots from the list whose startTime exists in the booked start times list.

6 **Apply Pagination**
Apply page and size to the remaining slots.

7 **Return Available Slots**
Return the final list of available slots.

STEP 2: Remove Booked Slots

Booked Start Times (for the Doctor & Date)

09:30 AM

10:30 AM

1 09:00 AM - 09:30 AM ✓

2 09:30 AM - 10:00 AM ✗

3 10:00 AM - 10:30 AM ✓

4 10:30 AM - 11:00 AM ✗

FINAL OUTPUT (Available Slots)

Slots after removing booked slots and applying pagination

1 09:00 AM - 09:30 AM
2 10:00 AM - 10:30 AM

(These slots are returned to the frontend)



SOURCE OF DATA

- Availabilities Table
(startTime, endTime, duration, day, doctorId)
- Appointments Table
(doctorId, date, startTime, status)
(status != CANCELLED)



EXAMPLE

Availability: 09:00 AM - 11:00 AM, Duration = 30 min
Booked Start Times: 09:30 AM, 10:30 AM
Generated Slots: 4
After Removing Booked: 2
Returned Slots: 2



KEY LOGIC

- Slots are generated in fixed intervals of duration.
- EndTime is exclusive (loop until $\text{endTime} - \text{duration}$).
- Only slots whose startTime is not in booked list (status != CANCELLED) are returned.

1. PATIENT MEDICAL RECORDS FLOW

Patient can manage Test Reports and view Past Consultations in Medical Records.

Patient Dashboard

- Dashboard
- Book Appointment
- My Appointments
- Prescriptions
- Medical Records

Medical Records

Patient clicks on Medical Records.

Two options will appear:

- Test Reports
- Past Consultations

A. TEST REPORTS (Clicked)

Patient can add, view, update and delete test reports.

Reports Table

+ Add Report

#	Report Name	Test Taken Date	File	Actions
1	Blood Test	01-06-2026		
2	X-Ray Chest	20-05-2026		
3	CBC Report	10-05-2026		

Add / Update Report (Form)

Report Name *

e.g. Blood Test

Test Taken Date *

01-06-2026

Upload File *

Choose File

No file chosen

SAVE REPORT

B. PAST CONSULTATIONS (Clicked)

Patient can view all previous consultations.

Consultations History List

#	Date	Doctor	Reason	Notes	View
1	01-06-2026	Dr. Smith	Fever & Cough	Patient had mild fever...	
2	20-05-2026	Dr. Smith	Chest Pain	Chest pain on left side...	
3	10-05-2026	Dr. John	Headache	Severe headache...	

Flow (Past Consultations)

1. Get patient from Principal
2. Get all appointments of patient
3. Get all consultations of those appointments
4. Return consultations list to frontend
5. Show in Consultations History List

2. DOCTOR VIEW MEDICAL RECORDS FLOW

During consultation, doctor can view patient's Test Reports and Past Consultations (view only).

A. DOCTOR - UPCOMING APPOINTMENTS

Upcoming Appointments

#	Date	Time	Patient Name	Actions
1	01-06-2026	09:00 AM	John Doe	Consultation
2	01-06-2026	09:30 AM	Mary Jane	Consultation
3	01-06-2026	10:00 AM	Robert Smith	Consultation

Doctor clicks "Consultation" for an appointment

B. CONSULTATION SCREEN

Appointment Details

Patient Name : John Doe
Date & Time : 01-06-2026, 09:00 AM
Doctor : Dr. Smith (Cardiologist)

[View Medical Records \(View Only\)](#)

Medical Records (View Only)

Test Reports (View Only)

Past Consultations (View Only)

Test Reports List (View Only)

#	Report Name	Test Taken Date	File
1	Blood Test	01-06-2026	
2	X-Ray Chest	20-05-2026	
3	CBC Report	10-05-2026	

Doctor can only view reports. (No Add / Update / Delete)

Past Consultations List (View Only)

#	Date	Doctor	Reason	Notes
1	01-06-2026	Dr. Smith	Fever & Cough	Patient had mild fever...
2	20-05-2026	Dr. Smith	Chest Pain	Chest pain on left side...
3	10-05-2026	Dr. John	Headache	Severe headache...

Doctor can only view consultations history. (No Add / Update / Delete)

C. FRONTEND REFLECTION



Doctor can now view patient's test reports and consultation history before filling consultation form and giving diagnosis.

D. BACKEND FLOW

1 Get Appointment by ID

Find appointment using appointmentId.

2 Extract Patient

Get patient from the appointment.

3 Fetch Patient's Reports

Fetch all reports where patientId = patient.id.

4 Fetch Patient's Consultations

Fetch all consultations of appointments where patientId = patient.id.

5 Return Reports & Consultations Separately

Return two lists separately to frontend.

Returned To Frontend (Two Separate DTOs)

TestReportDto (List)

```
{
  "id": 1,
  "reportName": "Blood Test",
  "testTakenDate": "01-06-2026",
  "fileUrl": "/files/blood-test.pdf"
} ...
```

PastConsultationDto (List)

```
{
  "id": 10,
  "date": "01-06-2026",
  "doctorName": "Dr. Smith",
  "reason": "Fever & Cough",
  "notes": "Patient had mild fever..."
} ...
```

DOCTOR PERMISSION

VIEW ONLY

- ✗ Add Report
- ✗ Update Report
- ✗ Delete Report

LEGEND: View Update Delete File Patient Doctor

Doctor Appointment System – Overall Flow (Frontend ↔ Backend)



PATIENT

Book Appointment, View Appointments, Profile, Consultation



DOCTOR

Manage Availability, Appointments, Add Consultation



ADMIN

Manage Doctors, View & Manage Appointments

START



1. REGISTER (Patient)

API: POST /api/auth/register



2. LOGIN

API: GET /api/auth/login



3. USER DETAILS

API: GET /api/auth/user-details



JWT Token

(Stored in Frontend)



1. PATIENT PROFILE (Create / Update / View)

Frontend Screen	Profile Form
API	POST /api/patient/add-or-update GET /api/patient/view-profile
Request (from)	Form Input (DOB, Gender, Contact) – (from token)
Response (from API)	Profile DTO
Shown in Frontend	Profile details shown / pre-filled in form



2. BOOK APPOINTMENT – Step 1 (Get Available Slots)

Frontend Screen	Select Doctor → Date → View Slots
API	GET /api/appointment/book-step1/{docId}
Request (from)	docId (from doctor list), page, size, appointmentDate (from date picker)
Response (from API)	List of Available Slots (startTime, endTime, duration)
Shown in Frontend	Slots displayed as clickable buttons/cards



3. BOOK APPOINTMENT – Step 2 (Confirm Booking)

Frontend Screen	Enter Symptoms, Reason → Confirm
API	POST /api/appointment/book-step2/{docId}
Request (from)	docId (from previous step), date, startTime, endTime, symptoms, reason (form)
Response (from API)	Success message
Shown in Frontend	"Appointment Booked Successfully" message



4. MY APPOINTMENTS (All / Filter / Cancel)

Frontend Screen	Table of Appointments + Filter by Status
APIs	GET /api/appointment/patient/all-appointments GET /api/appointment/patient/all-appointments-by-status PUT /api/appointment/patient-cancel-appointment/{appld}
Request (from)	page, size page, size, status (from dropdown) appld (from row action button)
Response (from API)	List of Appointment DTOs / Success
Shown in Frontend	Appointments in table (Status as badge), Cancel button for allowed status



5. VIEW CONSULTATION

Frontend Screen	Click "View Consultation" on Completed Appointment
API	GET /api/consultation/get-one/{appld}
Request (from)	appld (from row action button)
Response (from API)	Consultation DTO
Shown in Frontend	Consultation details page / modal



1. MANAGE AVAILABILITY (CRUD)

Frontend Screen	Weekly Availability Table
APIs	POST /api/avail/add GET /api/avail/view-availabilities PUT /api/avail/update/{id} DELETE /api/avail/delete/{id}
Request (from)	Form Input (day, startTime, endTime, duration) page, size id (from row), updated data id (from row)
Response (from API)	Success / List of Availabilities
Shown in Frontend	Availability table with Update / Delete buttons



2. APPOINTMENTS (Upcoming / Filter / Confirm / Cancel)

Frontend Screen	Upcoming Appointments Table + Actions
APIs	GET /api/appointment/doctor/all-upcoming-appointments GET /api/appointment/doctor/all-appointments-by-status PUT /api/appointment/confirm-appointment/{appld} PUT /api/appointment/doctor-cancel-appointment/{appld}
Request (from)	page, size page, size, status (from dropdown) appld (from row button) appld (from row button)
Response (from API)	List of Appointment DTOs / Success
Shown in Frontend	Appointments table sorted by date & time, Action buttons: Confirm, Cancel, Consultation



3. ADD CONSULTATION

Frontend Screen	Consultation Form (from Confirmed Appointment)
API	POST /api/consultation/add/{appld}
Request (from)	appld (from row button), symptoms, phyExam, treatment, recommended, prescription (form)
Response (from API)	Success (and appointment marked as COMPLETED)
Shown in Frontend	Success message, appointment status updated to COMPLETED



1. MANAGE DOCTORS (CRUD / View / Filter)

Frontend Screen	Doctors Table + Add / Update
APIs	POST /api/doctor/add GET /api/doctor/all GET /api/doctor/all-by-specialty GET /api/doctor/get-one/{id} PUT /api/doctor/update/{id}
Request (from)	Doctor form input page, size page, size, specialty (from dropdown) id (from row button) id (from row), updated data
Response (from API)	Success / List of Doctors / Doctor DTO
Shown in Frontend	Doctors table with View / Update buttons, Filter by specialty dropdown



2. APPOINTMENTS (By Doctor)

Frontend Screen	Select Doctor → View Appointments
APIs	GET /api/appointment/admin/all-upcoming-appointments/{docId} GET /api/appointment/admin/all-appointments-by-status/{docId} PUT /api/appointment/admin-cancel-appointment/{appld}/{docId}
Request (from)	docId, page, size docId, page, size, status (from dropdown) appld, docId (from row button)
Response (from API)	List of Appointment DTOs / Success
Shown in Frontend	Appointments table, Cancel button

LEGEND (Common for All)

Dropdowns / Selections	Doctor, Specialty, Status, Day, Date, Time, Gender
Row Actions	View, Update, Delete, Confirm, Cancel, Consultation
Status Badges	PENDING CONFIRMED CANCELLED COMPLETED
Pagination	All list APIs use page & size → show pagination controls
Auth	JWT Token sent in Authorization header for all secured APIs



FLOW END RESULT: Appointment Completed → Consultation Added → Patient can View Consultation